Session2-3.R

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```r
library(corrplot)
```

```
## corrplot 0.95 loaded
```

```r
## Set Working Directory
setwd("~/Desktop/MLWBA")
## Import Dataset
insurance = read.csv("insurance_LR1.csv")
summary(insurance)
```

```
##       age            sex                 bmi           children    
##  Min.   :18.00   Length:1338        Min.   :15.96   Min.   :0.000  
##  1st Qu.:27.00   Class :character   1st Qu.:26.27   1st Qu.:0.000  
##  Median :39.00   Mode  :character   Median :30.40   Median :1.000  
##  Mean   :39.21                      Mean   :30.66   Mean   :1.095  
##  3rd Qu.:51.00                      3rd Qu.:34.70   3rd Qu.:2.000  
##  Max.   :64.00                      Max.   :53.13   Max.   :5.000  
##                                     NA's   :2                      
##     smoker             region             charges     
##  Length:1338        Length:1338        Min.   : 1122  
##  Class :character   Class :character   1st Qu.: 4740  
##  Mode  :character   Mode  :character   Median : 9382  
##                                        Mean   :13270  
##                                        3rd Qu.:16640  
##                                        Max.   :63770  
## 
```

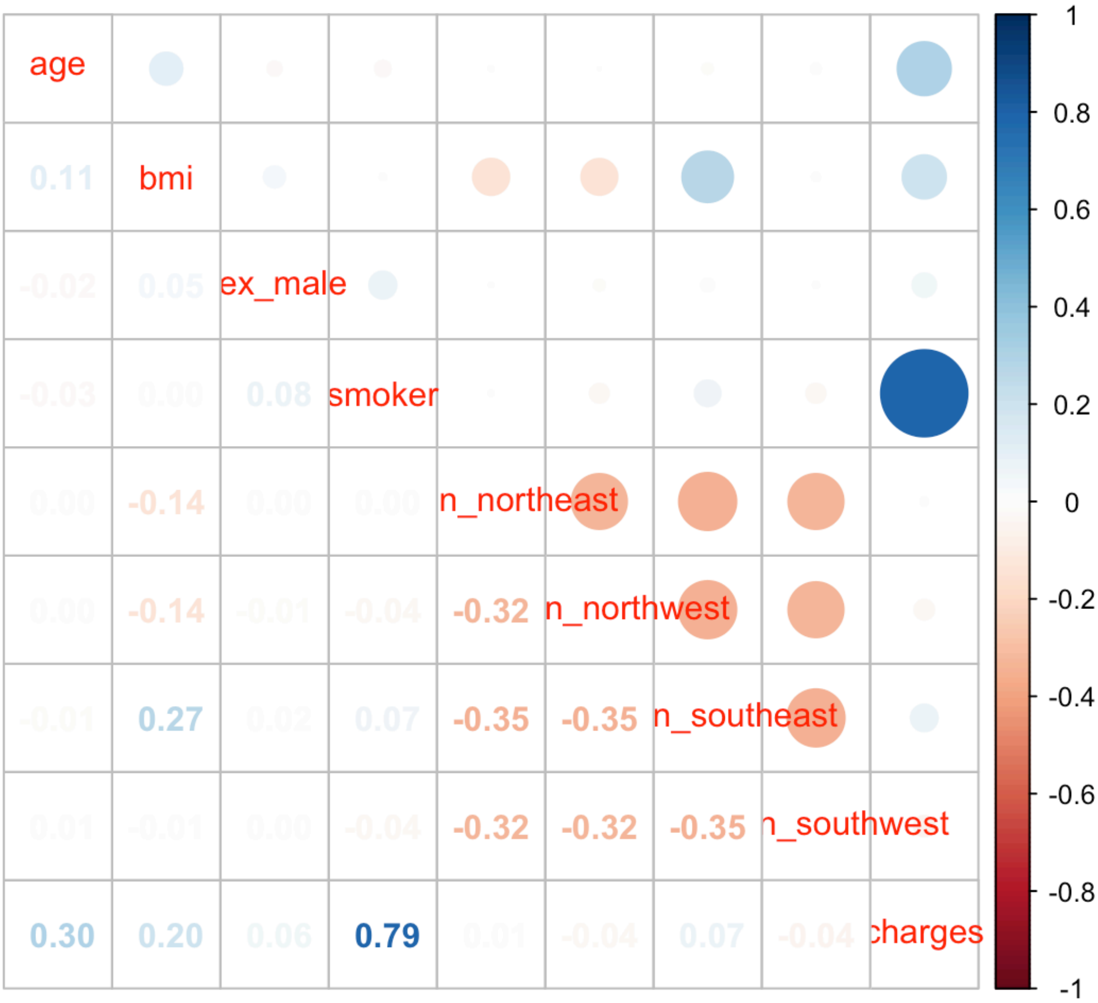
```r
View(insurance)
insurance = read.csv("insurance_LR1.csv",stringsAsFactors = TRUE)
View(insurance)
summary(insurance)
```

```
##       age             sex            bmi           children     smoker  
##  Min.   :18.00   female:662   Min.   :15.96   Min.   :0.000   no :1064  
##  1st Qu.:27.00   male  :676   1st Qu.:26.27   1st Qu.:0.000   yes: 274  
##  Median :39.00                Median :30.40   Median :1.000             
##  Mean   :39.21                Mean   :30.66   Mean   :1.095             
##  3rd Qu.:51.00                3rd Qu.:34.70   3rd Qu.:2.000             
##  Max.   :64.00                Max.   :53.13   Max.   :5.000             
##                               NA's   :2                                 
##        region       charges     
##  northeast:324   Min.   : 1122  
##  northwest:325   1st Qu.: 4740  
##  southeast:364   Median : 9382  
##  southwest:325   Mean   :13270  
##                  3rd Qu.:16640  
##                  Max.   :63770  
## 
```

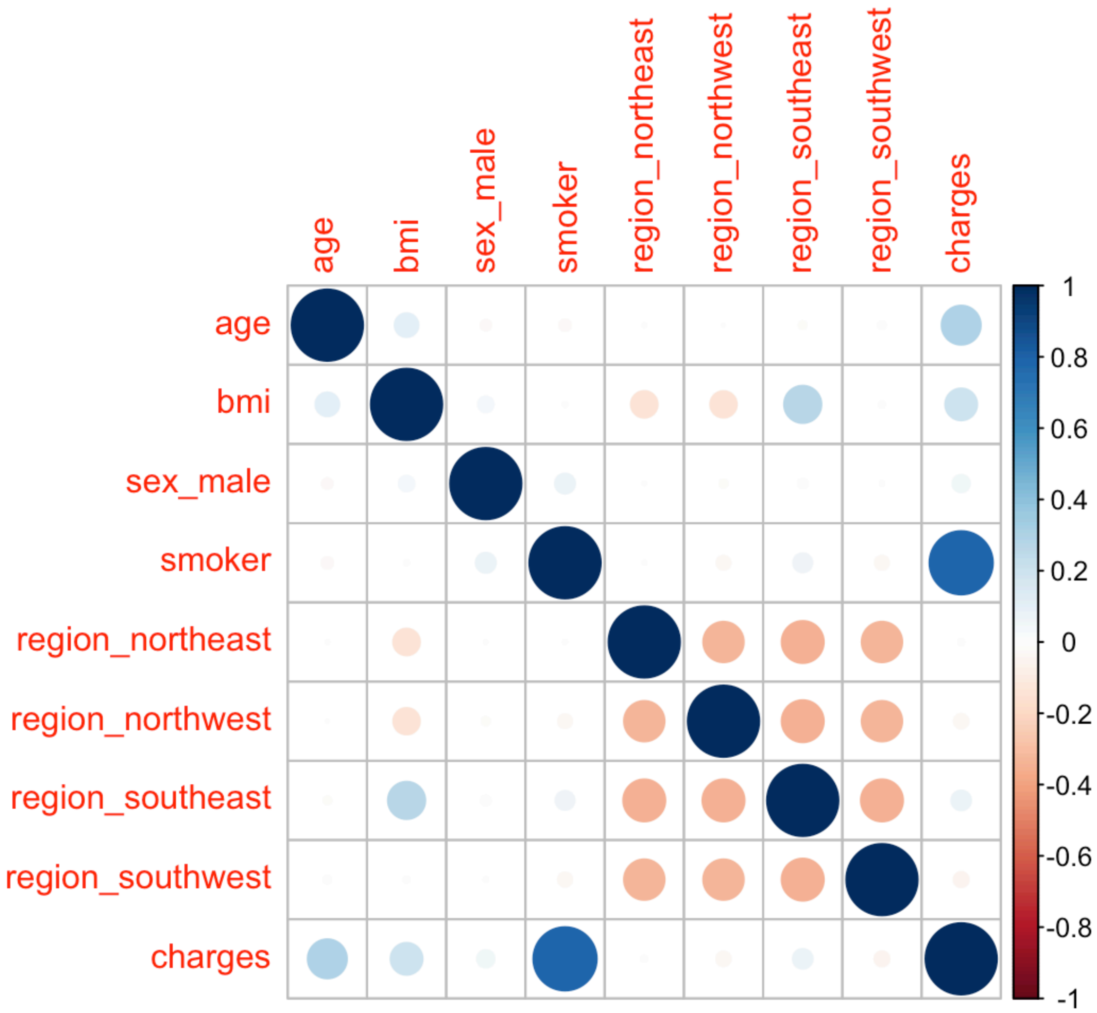
```
## convert into Categorical Data
insurance$children = as.factor(insurance$children)
insurance = na.omit(insurance)
## Step 2: Create Dummy Variable
## install.packages("fastDummies")
library(fastDummies)
insurance = dummy_cols(insurance, select_columns = c ("sex", "region"))
## label encoding
insurance$smoker = ifelse (insurance$smoker=="yes",1,0)
## Step 3: Correlation
cr = cor(insurance[c("age","bmi","sex_male","smoker","region_northeast","region_no
rthwest",
                     "region_southeast","region_southwest","charges")])
cr
```

```
##                           age          bmi     sex_male       smoker
## age                1.0000000000  0.109241926 -0.020534700 -0.025199587
## bmi                0.1092419260  1.000000000  0.046456676  0.003724879
## sex_male          -0.0205347004  0.046456676  1.000000000  0.077007886
## smoker            -0.0251995874  0.003724879  0.077007886  1.000000000
## region_northeast   0.0022825014 -0.138219181 -0.001589578  0.002382745
## region_northwest  -0.0006998183 -0.136261094 -0.012068244 -0.036544052
## region_southeast  -0.0110021671  0.270477280  0.016388733  0.068979499
## region_southwest   0.0098259531 -0.006236044 -0.003348203 -0.037391955
## charges            0.2988485844  0.198349821  0.058226416  0.787176783
##                   region_northeast region_northwest region_southeast
## age                    0.002282501    -0.0006998183      -0.01100217
## bmi                   -0.138219181    -0.1362610936       0.27047728
## sex_male              -0.001589578    -0.0120682444       0.01638873
## smoker                 0.002382745    -0.0365440519       0.06897950
## region_northeast       1.000000000    -0.3201581028      -0.34560430
## region_northwest      -0.320158103     1.0000000000      -0.34560430
## region_southeast      -0.345604303    -0.3456043031       1.00000000
## region_southwest      -0.320810336    -0.3208103356      -0.34630838
## charges                0.005851891    -0.0397312228       0.07482464
##                   region_southwest      charges
## age                    0.009825953  0.298848584
## bmi                   -0.006236044  0.198349821
## sex_male              -0.003348203  0.058226416
## smoker                -0.037391955  0.787176783
## region_northeast      -0.320810336  0.005851891
## region_northwest      -0.320810336 -0.039731223
## region_southeast      -0.346308375  0.074824639
## region_southwest       1.000000000 -0.043733841
## charges               -0.043733841  1.000000000
```

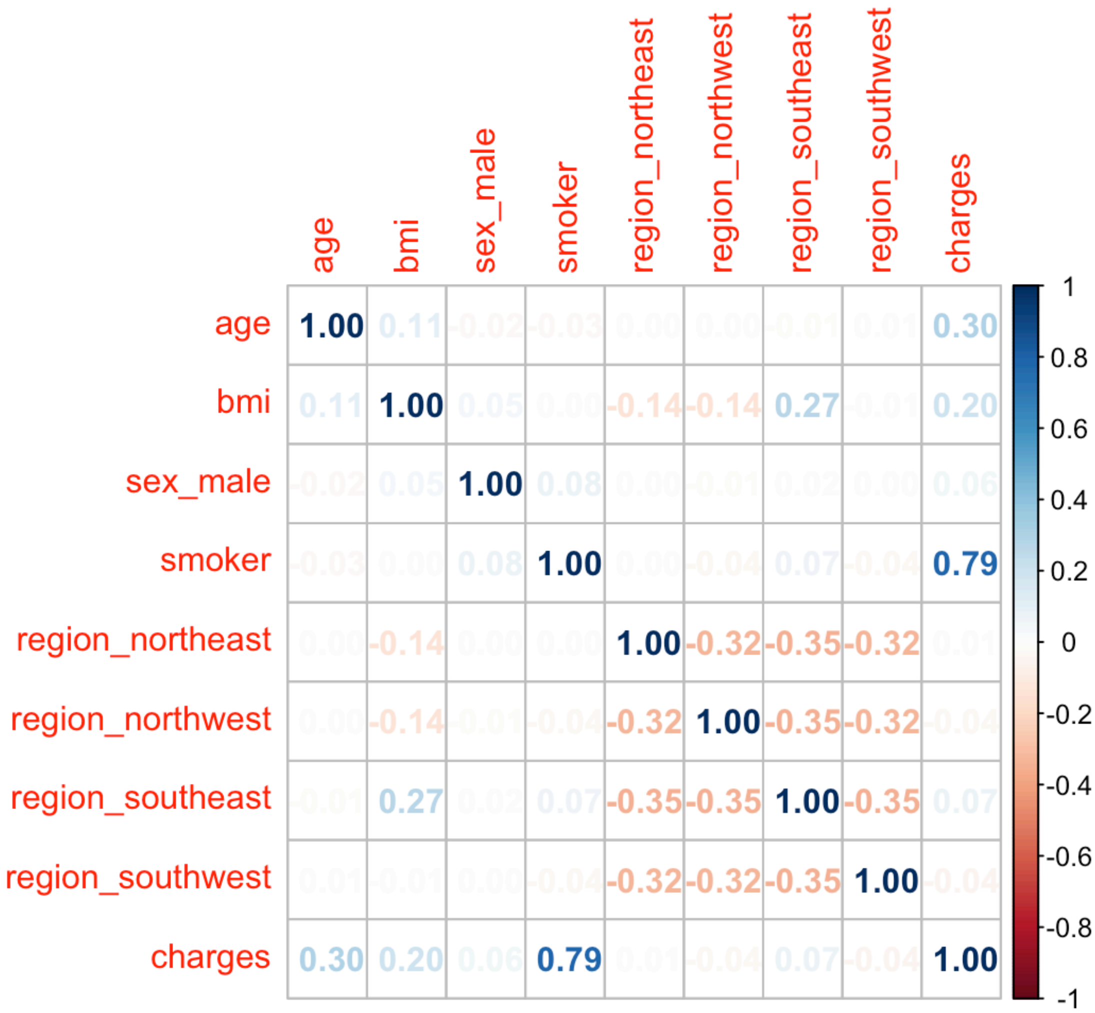
```
corrplot.mixed(cr)
```



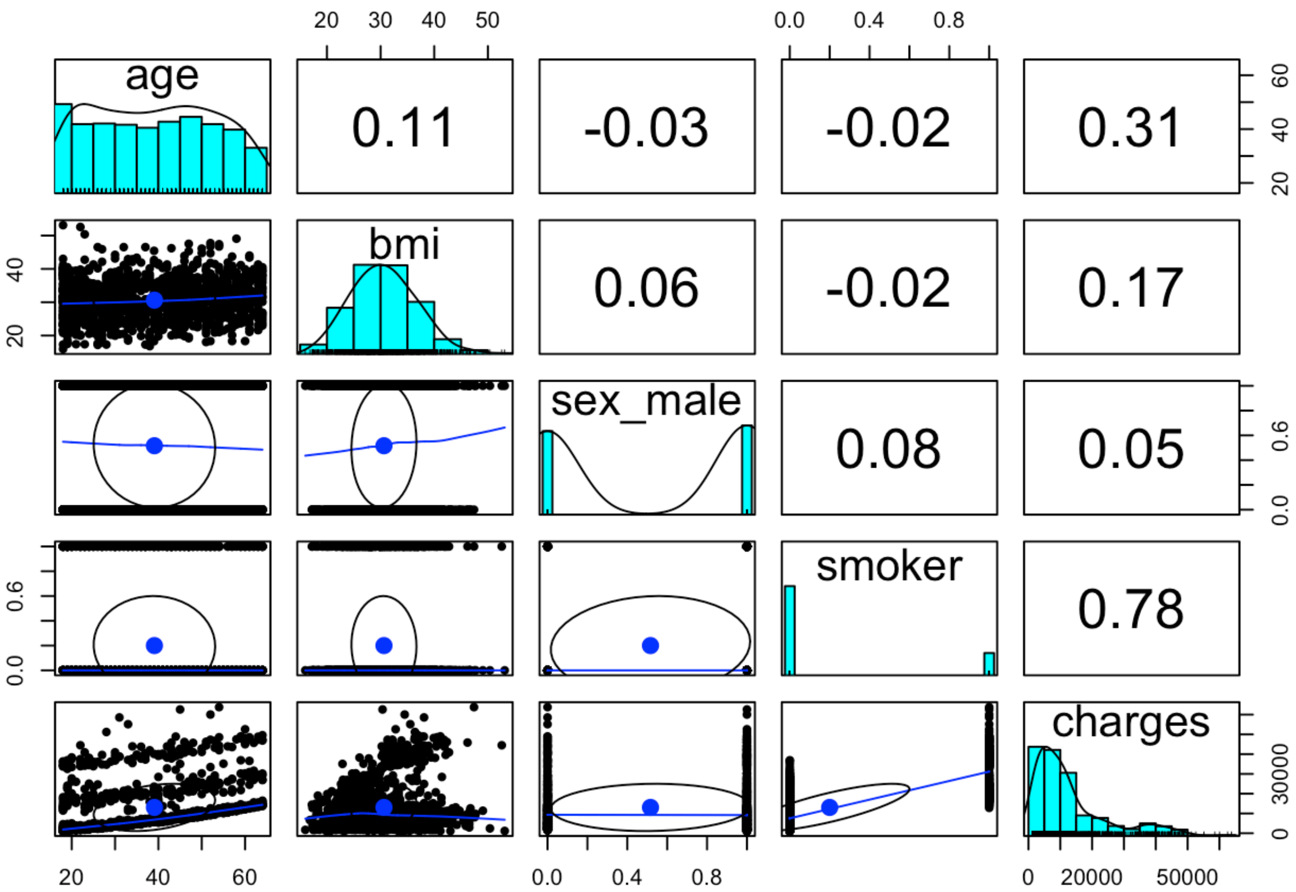

```
corrplot(cr,type = "full")
```




```
corrplot(cr,method = "number")
```




```
## help
?corrplot
## Step 4: Training & Testing (to do prediction)
## install.packages("caTools")
library(caTools)
set.seed(111)
split = sample.split(insurance$charges, SplitRatio = 0.85)
training_data = subset(insurance,split == "TRUE")
testing_data = subset(insurance,split == "FALSE")
View(training_data)
View(testing_data)
                    ## Note ##
## DV has to be normally distributed -> Hetero scarcity
## DV and IV must have correlation -> Insignificant Variable
## Two IV shouldn't be highly correlated -> Multicollinearity
library(psych)
pairs.panels(training_data[c("age","bmi","sex_male","smoker","charges")])
```




```
## Step 5:building the model on training dataset
model1 = lm(charges ~ age, data = training_data)
summary(model1)
```

```
##
## Call:
## lm(formula = charges ~ age, data = training_data)
##
## Residuals:
##    Min     1Q Median     3Q    Max
##  -7884  -6481  -5775   5344  47969
##
## Coefficients:
##             Estimate Std. Error t value Pr(>|t|)
## (Intercept)  2704.15    1000.28   2.703  0.00697 **
## age           264.89      24.09  10.994  < 2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 11380 on 1133 degrees of freedom
## Multiple R-squared:  0.09639,    Adjusted R-squared:  0.0956
## F-statistic: 120.9 on 1 and 1133 DF,  p-value: < 2.2e-16
```

```r
model2 = lm(charges ~ age+bmi, data = training_data)
summary(model2)
```

```
##
## Call:
## lm(formula = charges ~ age + bmi, data = training_data)
##
## Residuals:
##    Min     1Q Median     3Q    Max
## -12920  -6811  -5139   6559  48121
##
## Coefficients:
##             Estimate Std. Error t value Pr(>|t|)
## (Intercept) -5397.18    1877.57  -2.875  0.00412 **
## age           251.68      23.98  10.497  < 2e-16 ***
## bmi           281.40      55.42   5.077 4.48e-07 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 11250 on 1132 degrees of freedom
## Multiple R-squared:  0.1165, Adjusted R-squared:  0.115
## F-statistic: 74.64 on 2 and 1132 DF,  p-value: < 2.2e-16
```

```r
## vif = 1/(1-R^2) {if vif > 5 -> Multicollinearity}
library(car)
```

```
## Loading required package: carData
```

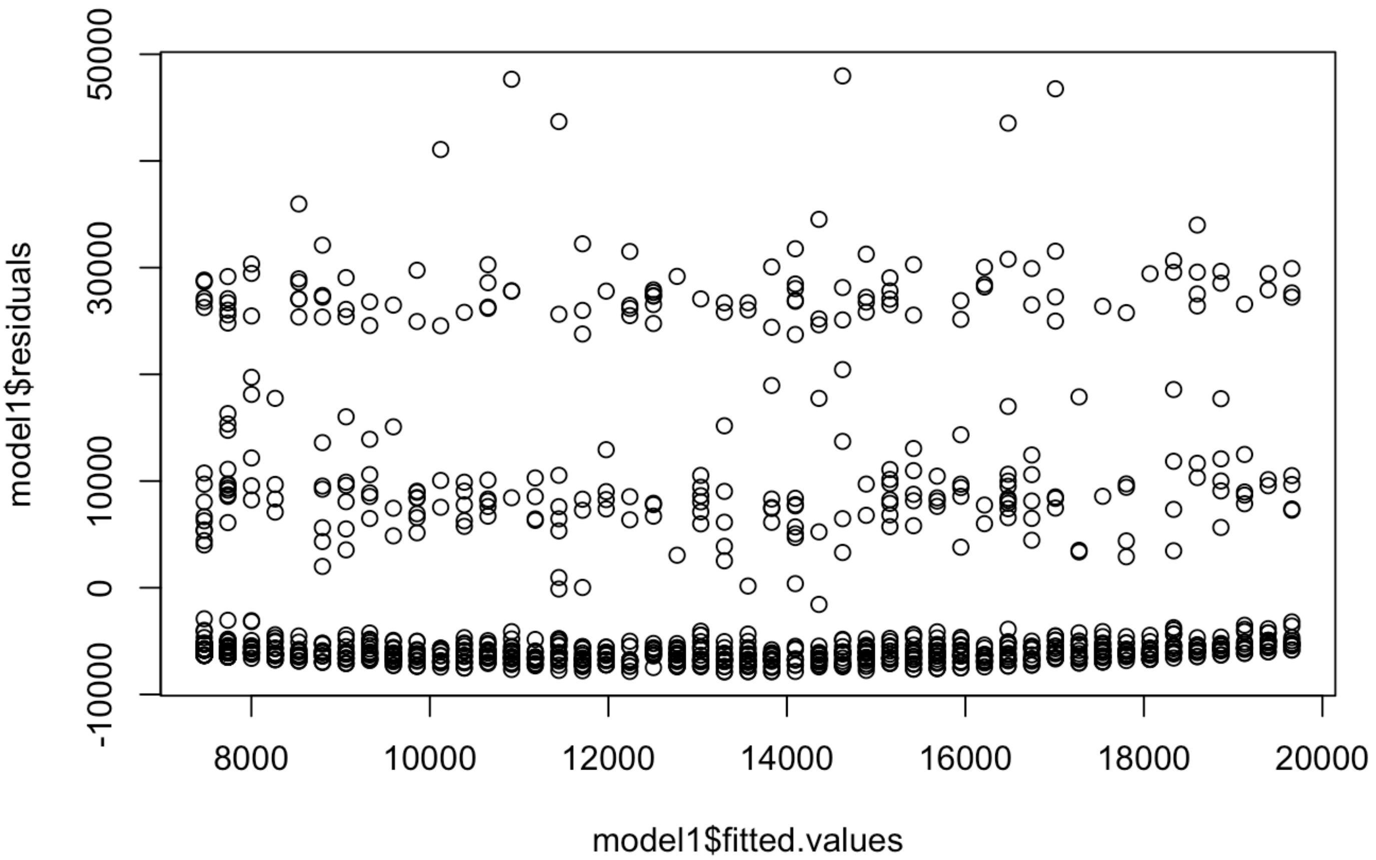
```
##
## Attaching package: 'car'
```

```
## The following object is masked from 'package:psych':
##
##     logit
```

```r
vif(model2)
```

```
##      age      bmi
## 1.011914 1.011914
```

```r
plot(model1$fitted.values,model1$residuals) ##funnal test
```




```
library(skedastic)
library(lmtest)
```

```
## Loading required package: zoo
```

```
##
## Attaching package: 'zoo'
```

```
## The following objects are masked from 'package:base':
##
##     as.Date, as.Date.numeric
```

```
lmtest::bptest(model1)
```

```
##
##  studentized Breusch-Pagan test
##
## data:  model1
## BP = 0.01957, df = 1, p-value = 0.8887
```

```
lmtest::bptest(model2)
```

```
##
##  studentized Breusch-Pagan test
##
## data:  model2
## BP = 95.796, df = 2, p-value < 2.2e-16
```

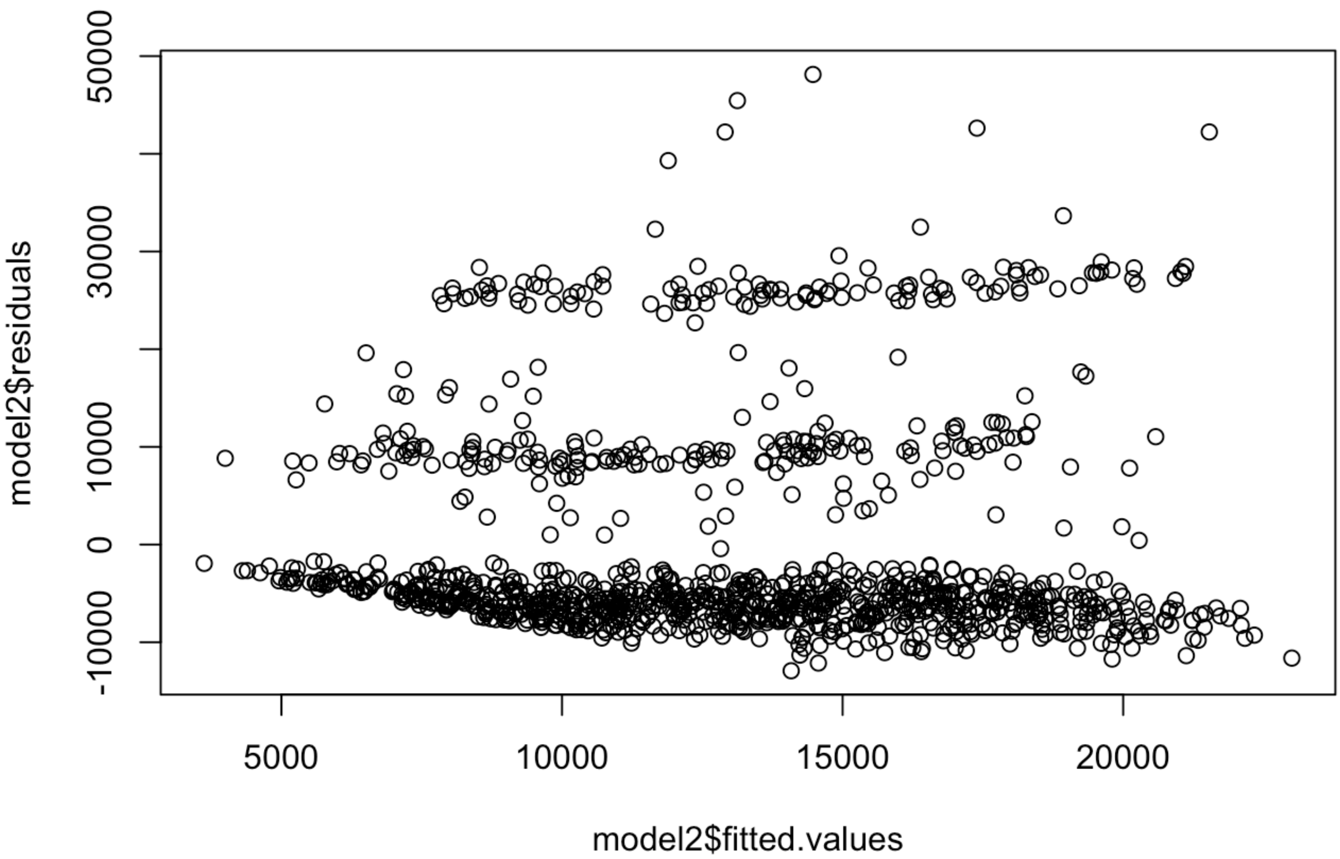
```
skedastic::white(model1)
```

```
## # A tibble: 1 × 5
##   statistic p.value parameter method       alternative
##       <dbl>   <dbl>     <dbl> <chr>        <chr>
## 1     0.583   0.747         2 White's Test greater
```

```
skedastic::white(model2) ##heteroscarcity -> doesn't want
```

```
## # A tibble: 1 × 5
##   statistic  p.value parameter method       alternative
##       <dbl>    <dbl>     <dbl> <chr>        <chr>
## 1      96.9 4.53e-20         4 White's Test greater
```

```
plot(model2$fitted.values,model2$residuals)
```



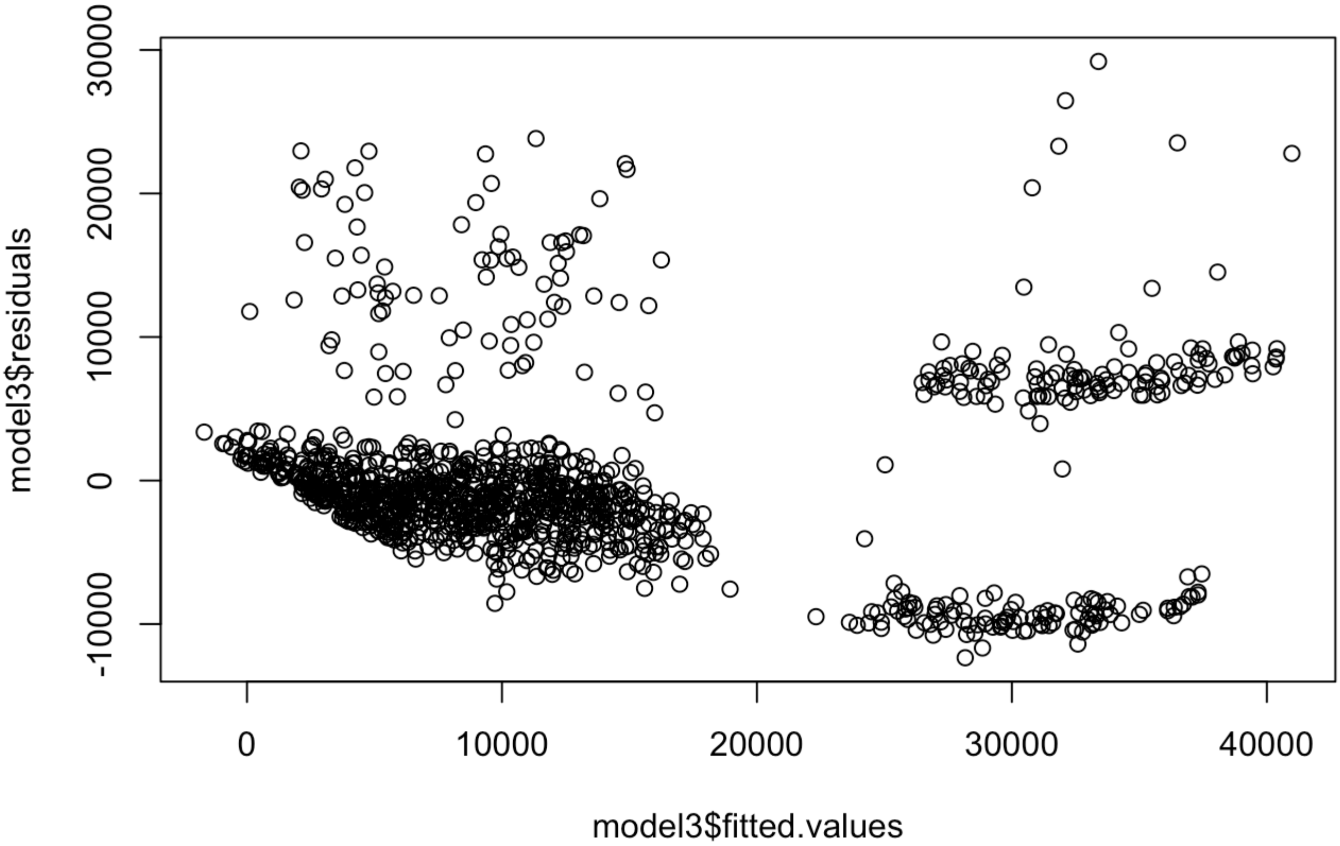
```
###
model3 = lm(charges ~ age+bmi+smoker, data = training_data)
summary(model3)
```

```
##
## Call:
## lm(formula = charges ~ age + bmi + smoker, data = training_data)
##
## Residuals:
##    Min     1Q Median     3Q    Max
## -12349  -2952  -1073   1316  29198
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept) -11285.68    1027.48  -10.98   <2e-16 ***
## age            261.79      13.04   20.07   <2e-16 ***
## bmi            306.73      30.15   10.17   <2e-16 ***
## smoker       23586.97     454.30   51.92   <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 6120 on 1131 degrees of freedom
## Multiple R-squared:  0.7389, Adjusted R-squared:  0.7382
## F-statistic:  1067 on 3 and 1131 DF,  p-value: < 2.2e-16
```

```
vif(model3)
```

```
##      age      bmi   smoker
## 1.012140 1.012179 1.000544
```

```
plot(model3$fitted.values,model3$residuals)
```




```
lmtest::bptest(model3)
```

```
##
##  studentized Breusch-Pagan test
##
## data:  model3
## BP = 101.65, df = 3, p-value < 2.2e-16
```

```
skedastic::white(model3)
```

```
## # A tibble: 1 × 5
##   statistic  p.value parameter method       alternative
##       <dbl>    <dbl>     <dbl> <chr>        <chr>
## 1      103. 6.89e-20         6 White's Test greater
```

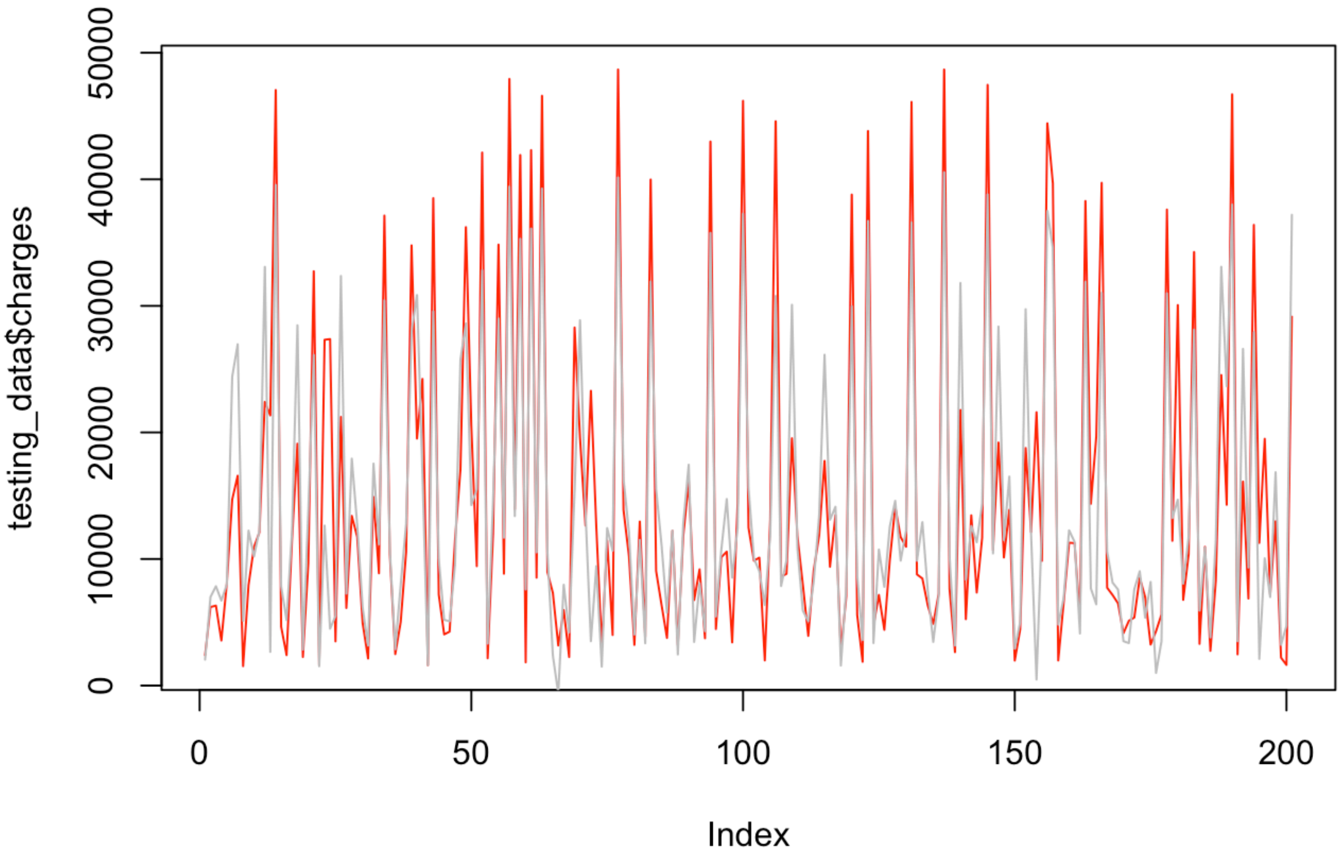
```
###

## test the model on testing data ##
prediction = predict(model3,testing_data)
head(prediction)
```

```
##        18         25         44         48         55         65
##  2049.637  6996.899  7848.084  6709.669  7986.259 24414.128
```

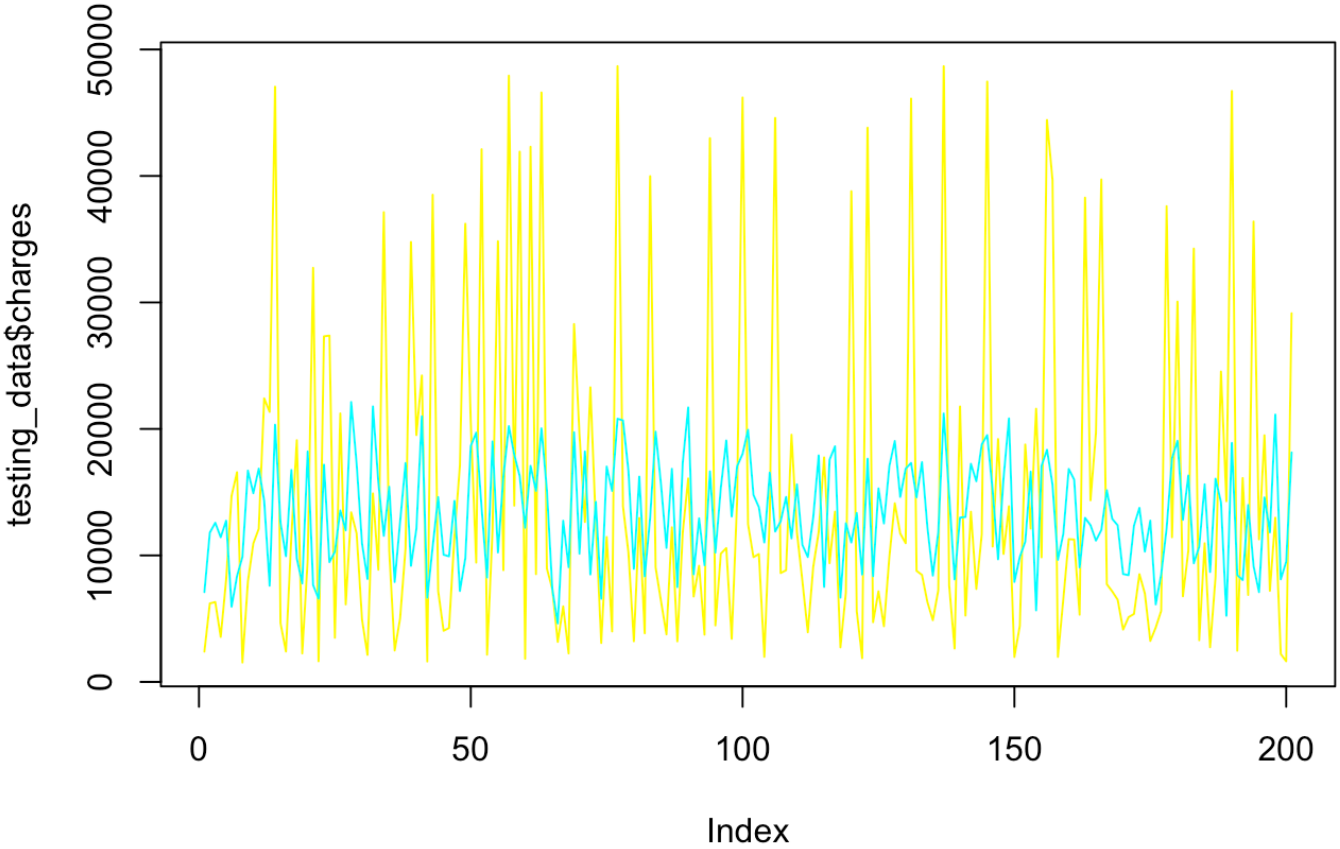
```
head(testing_data$charges)
```

```
## [1]  2395.172  6203.902  6313.759  3556.922  8059.679 14711.744
```

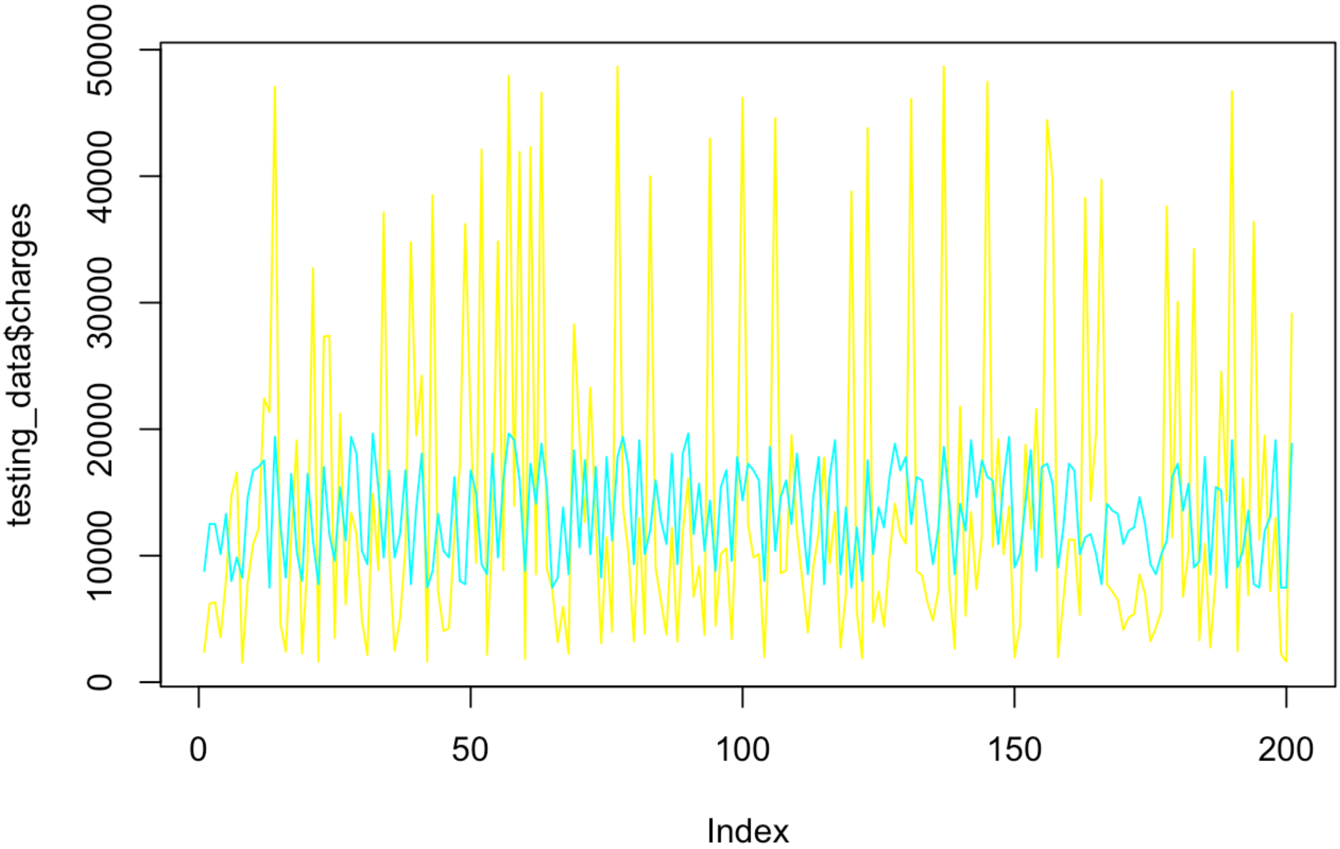
```
plot(testing_data$charges,type="l",col="red")
lines(prediction,type = "l", col="grey")
```




```
prediction1 = predict(model2,testing_data)
plot(testing_data$charges,type="l",col="yellow")
lines(prediction1,type = "l", col="cyan")
```




```
prediction2 = predict(model1,testing_data)
plot(testing_data$charges,type="l",col="yellow")
lines(prediction2,type = "l", col="cyan")
```



testing_data$charges
0
10000
20000
30000
40000
50000
0
50
100
150
200
Index